Nonlocal operators and Chaos

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Resumen: Caracterizamos el comportamiento caótico de operadores no locales que provienen de una amplia gama de esquemas numéricos temporales para operadores diferenciales fraccionarios. Para ello, utilizamos criterios de caos de operadores de Toeplitz en espacios de sucesiones de Lebesgue. Sorprendentemente, se demuestra que esta caracterización es (en algunos casos) dependiente del orden fraccionario del operador y del tamaño del paso del esquema numérico.

Abstract: We characterize the chaotic behavior of nonlocal operators that come from a broad class of time-stepping schemes of approximation for fractional differential operators. For that purpose, we use criteria for chaos of Toeplitz operators in Lebesgue spaces of sequences. Surprisingly, this characterization is proven to be (in some cases) dependent of the fractional order of the operator and the step size of the scheme.

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1. Introduction

In the last two decades, due to the introduction of fractional (or non-local) operators in the field of mathematical modeling of problems that arise in various areas of knowledge, many researchers have experienced a great interest in deeply understanding the behavior of these operators from different perspectives. One of the main efforts has been made to connect these operators with the modeling of partial differential equations as a new method to understand the processes that are closely related to having "memory" in their nature. On the other hand, it is well known that complex partial differential equation modeling needs an in-depth analysis of linear evolution systems.

The study of discrete nonlocal operators, having fractional order operators as a prototype, is a fashionable area of study nowadays. As it has been recently demonstrated, systems with this property can be used to develop data encryption algorithms. The basis for such a property is the chaotic behavior of the systems under consideration. With this motivation, we study necessary and sufficient conditions for chaos, uniquely based on the available data of nonlocal operators arising in numerical analysis, namely, the fractional order and the step size.

From an analytical perspective, the study of the chaotic behavior of nonlocal operators is an important first step that began recently to be studied by Edelman, Wu and Baleanu, among others. The reason is that, by their very nature, non-local operators seem naturally linked to a complex evolution. On the other hand, due to the high sensitivity of chaotic systems to parameters and initial conditions, chaos based algorithms are developed and studied as the core of encryption algorithms. For instance, many substitution encryption algorithms have been introduced based on discrete chaotic maps such as the conventional logistic and tent maps. This motivates the following questions: Is there any kind of chaotic behavior associated with the time approximation schemes for non-local (fractional) differential operators? If the answer is positive, is there any dependence on the chaos with respect to the fractional order and/or the size of the scheme step?

Lizama, Murillo and Peris solved the aforementioned questions in [10] by studying a broad class of nonlocal operators that include those given in the form of a finite convolution operator. They show the chaotic behavior of the classical fractional difference operator ∇_a^α and, surprisingly, they find the chaotic behaviour of some non-local operators associated to numerical schemes is dependent to the fractional order and the size of the scheme's time step, like the fractional explicit Euler scheme, the fractional backward Euler scheme, the fractional second order backward difference scheme and the fractional Crank-Nicolson scheme. We have collected the existing results about this topic which can be found in [10].

This paper is structured as follows. Section 2 is devoted to recall some preliminaries related to Complex Analysis, in particular some topics about Toeplitz operators, and two meaningful criteria. Section 3 introduces the notion of (Devaney) chaos and some definitions about operators related to fractional calculus. We also deal with the proof of an important transference principle that allows transferring results from the fractional derivative operator in the sense of Riemann-Liouville to the Nabla difference operator by means of conjugation by translation. Later, we prove, in Lebesgue spaces of sequences, the chaotic behavior of the fractional difference operator in the sense of Riemann-Liouville and, by the transference principle, we also obtain chaos for the Nabla fractional operator. For this purpose, firstly, we study a characterization of chaos for Toeplitz operators. Section 4 is devoted to studying the chaos for relevant nonlocal difference operators that arise after analysing fractional time-stepping numerical schemes.

2. Preliminaries

This section is devoted to introduce the notation and background material which will be needed in our later development. We will provide references for all the results that we mention here.

At some point we will relate the operators we will be working on to Toeplitz operators, which are well-known in Complex Analysis. For this reason, we are interested in studying some topics about Toeplitz operators.

In what follows, we denote by $\mathbb{D}$ (resp. $\overline{\mathbb{D}}$) the open (resp. closed) unit disk and put $\widehat{\mathbb{D}} = \mathbb{C} \setminus \mathbb{D}$, we also write $\mathbb{T}$ for the complex unit circle. Moreover, we denote by $L^2(\mathbb{T})$ the Hilbert space of (classes of)

complex-valued measurable square-integrable (with respect to Lebesgue measure) functions on $\mathbb{T}$ and by $L^\infty(\mathbb{T})$ the Banach space of all classes of measurable and essentially bounded functions on $\mathbb{T}$.

In order to introduce Toeplitz operators, two previous ingredients are needed. First of all, we realize that any bounded measurable function $g \in L^\infty(\mathbb{T})$ defines a multiplication operator $M_g : f \mapsto gf$ on $L^2(\mathbb{T})$. Indeed,

$$\|M_g f\|_{L^2(\mathbb{T})} = \left(\frac{1}{2\pi}\int_0^{2\pi} |g(e^{i\theta})f(e^{i\theta})|^2 d\theta\right)^{1/2} \leq \|g\|_{L^\infty(\mathbb{T})}\|f\|_{L^2(\mathbb{T})}, \qquad \text{for all } f \in L^2(\mathbb{T}).$$

Secondly, we must mention the **Hardy space** of holomorphic functions on the unit disk. It is defined as

$$H^2(\mathbb{D}) = \left\{ f \in H(\mathbb{D}) : \|f\| := \sup_{0<r<1} \left(\frac{1}{2\pi}\int_0^{2\pi} |f(re^{i\theta})|^2 d\theta\right)^{1/2} < \infty \right\}.$$

An essential fact is that $H^2(\mathbb{D})$ is a closed subspace of $L^2(\mathbb{T})$, therefore the projection exists and it is well-defined due to $L^2(\mathbb{T})$ being a Hilbert space. Let $P : L^2(\mathbb{T}) \to H^2(\mathbb{D})$ be the projection. It is also known that $\{z^n : n \geq 0\}$ is an orthogonal basis in $H^2(\mathbb{D})$ (see [3]).
Now we have the tools to define Toeplitz operators.

Definition 1. Given a function $g \in L^\infty(\mathbb{T})$, the operator $T_g : L^2(\mathbb{T}) \to H^2(\mathbb{D})$ is defined as

$$T_g = P \circ M_g.$$

T_g is called a **Toeplitz operator** with symbol g. ◂

If we write $g(z) = \sum_{n\in\mathbb{Z}} a_n z^n \in L^\infty(\mathbb{T})$, given $f(z) = \sum_{n=0}^{\infty} b_n z^n \in H^2(\mathbb{D})$, we have

$$(T_g f)(z) = \sum_{n=0}^{\infty} c_n z^n \in H^2(\mathbb{D}),$$

where the sequence $c = (c_n)_n$ is obtained as the convolution of $a = (a_n)_n$ with $b = (b_n)_n$; i.e., $c = a * b$, defined as

$$c_n = (a * b)_n = \sum_{j=-\infty}^{n} a_j b_{n-j}, \qquad n \in \mathbb{N}_0 = \mathbb{N} \cup \{0\}.$$

In case that $g(z) = \sum_{n\in\mathbb{Z}} a_n z^n$ is so that $a = (a_n)_n \in \ell^1(\mathbb{N}_0)$, then T_g is a well-defined bounded operator on $\ell^2(\mathbb{N}_0)$.

The following result gives an useful characterization of these operators.

Proposition 2. *A bounded operator on $H^2(\mathbb{D})$ is Toeplitz if and only if its matrix representation in the basis $\{z^n : n \geq 0\}$ has constant diagonals.*

For further use, we recall the next interesting results. The first one was obtained in [2] and it provides "chaos" (in the following section we explain the meaning of chaos) for the kind of Toepliz operators that we are interested in. In the original paper [2], the authors only showed hypercyclicity for such operators, but chaos was easily reached in [10].

Theorem 3 (Baranov and Lishanskii (2016) and Lizama, Murillo and Peris (2020)). *Let $\Phi(z) = \frac{\gamma}{z} + \varphi(z)$ with $\gamma \in \mathbb{C}$ and $\varphi \in A(\overline{\mathbb{D}}) = H^\infty(\mathbb{D}) \cap C(\overline{\mathbb{D}})$ satisfying*

(i) the function Φ is univalent (i.e., injective) in $\overline{\mathbb{D}}\backslash\{0\}$;

(ii) $\mathbb{D} \cap (\mathbb{C}\backslash\Phi(\mathbb{D})) \neq \emptyset$ and $\widehat{\mathbb{D}} \cap (\mathbb{C}\backslash\Phi(\mathbb{D})) \neq \emptyset$.

Then the Toeplitz operator $T_\Phi : \ell^2(\mathbb{N}_0) \to \ell^2(\mathbb{N}_0)$ is Devaney chaotic.

The following theorem is an useful criterion due to M. R. Ganigi and B. A. Uralegaddi [5] that ensures univalence for meromorphic functions.

Theorem 4 (Ganigi and Uralegaddi, 1989). *Let M_n denote the class of functions of the form $f(z) = \frac{1}{z} + \sum_{n=0}^{\infty} a_n z^n$ which are regular in $0 < |z| < 1$ and satisfy*

$$\Re\left(\frac{D^{n+1}f(z)}{D^n f(z)} - 2\right) < -\frac{n}{n+1}, \quad \text{for } |z| < 1,$$

where $D^n f(z) = \frac{1}{z}(z^{n+1}\frac{f(z)}{n!})^{(n)}$, $n \in \mathbb{N}_0$. Then $M_{n+1} \subset M_n$ for all $n \in \mathbb{N}_0$ and all functions in M_n are univalent.

The last result we want to mention is the well-known Young's convolution generalized inequality [12], that will be helpful later.

Theorem 5 (Young's Theorem, 1912). *Suppose f is in $L^p(\mathbb{R}^N)$ and $g \in L^q(\mathbb{R}^N)$ and $1/p + 1/q = 1/r + 1$ with $1 \leq p, q,\ r \leq \infty$. Then $f * g \in L^r(\mathbb{R})^N$ and the following estimate holds*

$$\|f * g\|_r \leq \|f\|_p \|g\|_q.$$

3. Chaos of nonlocal operators

Definition 6 (Hypercyclicity). Consider X a Banach space and an operator $T : X \to X$. T is called **hypercyclic** if there exists a vector $x \in X$ such that $Orb(x, T) := \{T^n x : n \in \mathbb{N}\}$ is dense in X. ◂

Additionally, we define the following property of the operator T. It encapsulates one of the the main characteristics we expect from a "chaotic" operator to have, namely, that iterating T must separate points of X arbitrarily close to each other.

Definition 7. T is topologically transitive if for any pair of nonempty sets $U, V \subset X$ there exists $n \in \mathbb{N}$ such that $T^n(U) \cap V \neq \emptyset$. ◂

Theorem 8 (Birkhoff transitivity theorem). *Let T be a continuous map on a separable complete metric space X without isolated points. Then T is hypercyclic if and only if it is topologically transitive.*

Definition 9. T is said to have **sensitive dependence on initial conditions** if there exists some $\delta > 0$ such that, for every $x \in X$ and $\varepsilon > 0$, there exists some $y \in X$ with $d(x, y) < \varepsilon$ such that, for some $n \geq 0$, $d(T^n x, T^n y) > \delta$. ◂

Given an operator T in a Banach space, it is sufficient to ensure hypercyclicity to obtain sensitive dependence on initial condition thanks to the following result [7].

Theorem 10 (Banks, Brooks, Cairns, Davis and Stacey). *Let X be a metric space without isolated points. If a dynamical system $T : X \to X$ is topologically transitive and has a dense set of periodic points then T has sensitive dependence on initial conditions with respect to any metric defining the topology of X.*

We now define the main property of the operators we will study in this article. The idea is to define what we mean by a "chaotic operator". Clearly, we expect T to be hypercyclic and have sensitive dependence on initial conditions. Note that by Theorem 10, only hypercyclicity must be imposed. We also require the set of periodic points, $Per(T) := \{x \in X : T^n x = x \text{ for some } n\}$, to be dense in X. That is, we have the following fundamental definition.

Definition 11. An operator $T : X \to X$ is called **chaotic in the sense of Devaney** if it is hypercyclic, and $Per(T)$ is dense in X. ◂

We restrict ourselves to the above definition of chaos, but note that other definitions besides Devaney chaos exist (see, for example, [4]).

We are now in a position to exhibit the first example of a Devaney chaotic operator. Consider the sequence space ℓ^p, with $1 \leq p < \infty$, and the backward shift operator $B : \ell^p \to \ell^p$ defined as

$$(x_1, x_2, \dots) \mapsto (x_2, x_3, \dots). \tag{1}$$

If we consider now $|\lambda| > 1$, Rolewicz showed in 1969 that the operator λB is Devaney chaotic (see [11]).

Our next task is to characterize when two operators have equivalent dynamics. More precisely, we have the following definition.

Definition 12. Given two operators T and S defined on Banach spaces X and Y, respectively, we say T is **quasi-conjugate** to S if there exists a continuous map $\Phi : Y \to X$ with dense range such that $T \circ \Phi = \Phi \circ S$. ◂

Frequently, when dealing with two operators, one has information about the dynamics of one of them, and would like to extract information about the other. Definition 12 is then specially useful, since quasi-conjugation preserves most of the usual notions of linear dynamics. Specifically, hypercyclicity and Devaney chaos are preserved ([7], Proposition 1.31). We will later use this fact to prove an operator is Devaney chaotic, knowing it is quasi-conjugated to a Devaney chaotic one.

The operators we are going to consider are related to fractional calculus. Fractional calculus studies differential operators of arbitrary real order, thus generalizing differential operators of integer order. Fractional operators are particularly interesting, since they are non-local and incorporate memory effects into modelling. This means that fractional operators are an effective tool to study natural processes where heredity plays an important role, for example, in chemistry, control theory, statistics,...

Let us add a brief comment on the history of fractional calculus. It was in 1956 when Kutter mentioned by the first time differences of fractional order. Later, in 1974, Diaz and Osler described fractional operators as infinite series. Fifteen years later, Miller and Rose studied fractional sums via the solutions to linear difference equations. Recently, in 2007, Atici and Eloe introduced the Riemann-Liouville like fractional difference using the fractional sum of Miller and Rose. One of our objectives is to analyze the chaotic dynamic of the fractional difference operator defined in the sense of Riemann-Liouville.

In order to introduce the fractional difference operator (in the sense of Riemann-Liouville), we first need the following definitions. For a given $a \in \mathbb{N}$, we will denote $\mathbb{N}_a := \{a, a+1, a+2, \dots\}$, and $s(\mathbb{N}_a)$ the vector space of complex-valued sequences in $\mathbb{N}_a$. The following operator will be used throughout this article.

Definition 13 (Forward Euler operator). The Forward Euler operator $\Delta_a : s(\mathbb{N}_a) \to s(\mathbb{N}_a)$ is defined by

$$\Delta_a f(n) := f(n+1) - f(n), \quad n \in \mathbb{N}_a.$$

◂

Now, Δ_a^m is defined recursively as $\Delta_a^m : s(\mathbb{N}_a) \to s(\mathbb{N}_a)$ by $\Delta_a^m := \Delta_a^{m-1} \circ \Delta_a$. We will refer to it as the m-th order forward operator. When $a = 0$, we write $\Delta \equiv \Delta_0$, and we will the denote the 0-th power $\Delta_a^0 \equiv I_a$, the identity operator in $s(\mathbb{N}_a)$.

Using the definition of Δ_a, it is easy to prove by induction that Δ_a^m acts on a sequence $f \in s(\mathbb{N}_0)$ according to the formula

$$\Delta^m f(n) = \sum_{j=0}^{m} \binom{m}{j} (-1)^{m-j} f(n+j), \quad n \in \mathbb{N}_0. \tag{2}$$

Next, we define the following sequence for purely technical reasons. It will come in handy when manipulating, for instance, fractional sums.

Definition 14. For $\alpha \in \mathbb{R} \setminus \{0\}$, we denote

$$k^\alpha(n) = \begin{cases} \dfrac{\alpha(\alpha+1)\dots(\alpha+n-1)}{n!} & n \in \mathbb{N}_0, \\ 0 & otherwise. \end{cases}$$

◂

The above definition is extended for $\alpha = 0$ by writing $k^0(n) = 1$ if $n = 0$ and 0 otherwise. Observe that if $\alpha \in \mathbb{R} \setminus \{-1, -2, ..\}$, we can write k^α in terms of the Euler gamma function Γ as follows:

$$k^\alpha(n) = \frac{\Gamma(\alpha + n)}{\Gamma(\alpha)\Gamma(n+1)}, \ n \in \mathbb{N}_0. \tag{3}$$

As a consequence, one can use the known recursive properties of Γ to derive recursive relations for the sequence k^α.

Throughout this article, we will constantly use the following definitions. Note that they constitute generalizations of the integer cases, and the analogy with their continuous counterparts is apparent.

Definition 15 (Atici and Eloe (2009), [1]). Let $\alpha > 0$. For any given positive real number a, the α **-th fractional sum** of a function f is

$$\nabla_a^{-\alpha} f(n) := \sum_{j=a}^{n} k^\alpha(n-j) f(j).$$

◂

Definition 16. For each $\alpha > 0$ and a sequence $f \in s(\mathbb{N}_0)$, we define the α-th fractional sum

$$\Delta^{-\alpha} f(n) := (k^\alpha * f)(n) = \sum_{j=0}^{n} k^\alpha(n-j) f(j), \quad n \in \mathbb{N}_0.$$

◂

Definition 17. The **fractional difference operator** $\Delta^\alpha : s(\mathbb{N}_0) \to s(\mathbb{N}_0)$ of order $\alpha > 0$ in the sense of Riemann-Liouville is defined by

$$\Delta^\alpha f(n) := \Delta^m \circ \Delta^{-(m-\alpha)} f(n), \quad n \in \mathbb{N}_0,$$

where $m - 1 < \alpha < m$, $m := \lceil \alpha \rceil$. ◂

In particular, a straightforward calculation shows that for every $n \in \mathbb{N}_0$,

$$\Delta^\alpha u(n) = \Delta(k^{1-\alpha} * u)(n) = \sum_{j=0}^{n+1} k^{1-\alpha}(n+1-j) u(j) - \sum_{j=0}^{n} k^{1-\alpha}(n-j) u(j). \tag{4}$$

Definition 18. The **nabla fractional difference operator** $\nabla^\alpha : s(\mathbb{N}_a) \to s(\mathbb{N}_a)$ of order $\alpha > 0$ is defined by

$$\nabla_a^\alpha f(t) = \Delta_a^m \circ \nabla_a^{-(m-\alpha)} f(t), \quad t \in \mathbb{N}_a, \tag{5}$$

where $m - 1 < \alpha < m$, $m = \lceil \alpha \rceil$. ◂

The following theorem, called the *transference principle*, shows that ∇_a^α and Δ^α are quasi-conjugate (note that τ_α is clearly continuous and bijective). As we mentioned after Definion 12, this means that both are simultaneously Devaney chaotic or not Devaney chaotic.

Theorem 19 (Transference principle, [10]). *Let $\alpha > 0$ and $a \in \mathbb{R}$ be given. Then we have*

$$\tau_a \circ \nabla_a^\alpha = \Delta^\alpha \circ \tau_a,$$

where $\tau_a : s(\mathbb{N}_a) \to s(\mathbb{N}_0)$ by $\tau_a g(n) := g(a+n), \quad n \in \mathbb{N}_0$.

Proof. First, we compute $\tau_a \circ \nabla_a^{-\alpha}$:

$$
\begin{aligned}
\tau_a \circ \nabla_a^{-\alpha} f(n) &= \nabla_a^{-\alpha} f(n+a) = \sum_{s=a}^{n+a} k^{\alpha}(n+a-s) f(s) \\
&\underset{j:=s-a}{=} \sum_{j=0}^{n} k^{\alpha}(n-j) f(j+a) \\
&= \sum_{j=0}^{n} k^{\alpha}(n-j) \tau_a f(j) \\
&= (k^{\alpha} * \tau_a f)(n) = \Delta^{-\alpha} \circ \tau_a f(n).
\end{aligned} \tag{6}
$$

Now we use this result to end the proof of the transference principle:

$$
\begin{aligned}
\tau_a \circ \nabla_a^{\alpha} f(n) &= (\tau_a \circ (\Delta_a^m \circ \nabla_a^{-(m-\alpha)}) f)(n) = (\Delta_a^m \circ \nabla_a^{-(m-\alpha)} f)(n+a) \\
&= \sum_{j=0}^{m} \binom{m}{j} (-1)^{m-j} (\nabla_a^{-(m-\alpha)} f)(n+a+j) \\
&= \sum_{j=0}^{m} \binom{m}{j} (-1)^{m-j} (\tau_a \circ \nabla_a^{-(m-\alpha)} f)(n+j) \\
&\underset{(6)}{=} \sum_{j=0}^{m} \binom{m}{j} (-1)^{m-j} (\Delta^{-(m-\alpha)} \circ \tau_a f)(n+j) \\
&= \Delta^m (\Delta^{-(m-\alpha)} \circ \tau_a f)(n) = \Delta^{\alpha} \circ \tau_a f(n).
\end{aligned}
$$

■

The following result shows that the fractional difference operator is, indeed, a Toeplitz operator. As a consequence we could use the criterion for this kind of operators to study the chaotic behavior of Δ^{α}.

Lemma 20 (Matrix representation of Δ^{α}, [10]). *The representation of Δ^{α} in the canonical basis $\{e_l(j)\}_{j,l\in\mathbb{N}_0}$ of $\ell^2(\mathbb{N}_0)$ is a **Toeplitz matrix**. For $0 < \alpha < 1$ we have*

$$
\Delta^{\alpha} e_l(n) = \begin{cases} -\alpha \dfrac{k^{1-\alpha}(n-l)}{n-l+1} & \text{if } n \geq l \\[2ex] 1 & \text{if } n = l-1 \\[2ex] 0 & \text{if } n < l-1. \end{cases}
$$

The symbol of Δ^{α} as a Toeplitz operator is $\Phi(z) = \frac{(1-z)^{\alpha}}{z}$.

Proof. Our goal is to use the characterization given in Proposition 2. For that purpose, we need the matrix representation of Δ^{α}. Firstly, we compute Δ^{α} for the elements of the canonical basis. It is important to remember that $e_m(j)$ denotes the Kronecker delta, i.e.,

$$
e_m(j) = \begin{cases} 1 & \text{if } m = j, \\ 0 & \text{if } m \neq j. \end{cases}
$$

Fix $n \in \mathbb{N}_0$ and let $l \in \mathbb{N}_0$, we distinguish three cases:

- Case $l \leq n$:

$$\begin{aligned}
\Delta^\alpha e_l(n) &\underset{(4)}{=} \sum_{j=0}^{n+1} k^{1-\alpha}(n+1-j)e_l(j) - \sum_{j=0}^{n} k^{1-\alpha}(n-j)e_l(j) \\
&= k^{1-\alpha}(n+1-l) - k^{1-\alpha}(n-l) = k^{1-\alpha}(n-l)\left(\frac{1-\alpha+n-l}{n-l+1} - 1\right) \\
&= -\alpha\frac{k^{1-\alpha}(n-l)}{n-l+1},
\end{aligned}$$

  where in the third equality we have used the next property of the kernels: $k^\alpha(n+1) = \dfrac{\alpha+n}{n+1}k^\alpha(n)$ for $\alpha > 0$, $n \in \mathbb{N}_0$ ([6], Proposition 3.1).

- Case $l = n+1$:

$$\Delta^\alpha e_l(n) \underset{(4)}{=} \sum_{j=0}^{n+1} k^{1-\alpha}(n+1-j)e_l(j) - \sum_{j=0}^{n} k^{1-\alpha}(n-j)e_l(j) = k^{1-\alpha}(0) = 1,$$

  because $e_l(j) = e_{n+1}(j) = 0$ if $j \neq n+1$.

- Case $l > n+1$:

$$\Delta^\alpha e_l(n) \underset{(4)}{=} \sum_{j=0}^{n+1} k^{1-\alpha}(n+1-j)e_l(j) - \sum_{j=0}^{n} k^{1-\alpha}(n-j)e_l(j) = 0,$$

  for the same reason than in the previous case.

To sum up, the matrix representation of Δ^α in the canonical basis $\{e_l(j)\}_{j,l\in\mathbb{N}_0}$ is given by

$$\begin{bmatrix}
-\alpha & 1 & 0 & 0 & \dots \\
-\alpha\frac{k^{1-\alpha}(1)}{2} & -\alpha & 1 & 0 & \dots \\
-\alpha\frac{k^{1-\alpha}(2)}{3} & -\alpha\frac{k^{1-\alpha}(1)}{2} & -\alpha & 1 & \dots \\
-\alpha\frac{k^{1-\alpha}(3)}{4} & -\alpha\frac{k^{1-\alpha}(2)}{3} & -\alpha\frac{k^{1-\alpha}(1)}{2} & -\alpha & \dots \\
\vdots & \vdots & \vdots & \vdots & \ddots
\end{bmatrix}$$

We notice that the matrix has constant diagonals, therefore Proposition 2 ensures that it is a Toeplitz matrix. Consequently, remembering that $\{z^n : n \in \mathbb{Z}\}$ is an orthogonal basis in $L^2(\mathbb{T})$ (see [3]), the symbol of Δ^α as a Toeplitz operator is:

$$\begin{aligned}
\Phi(z) &= \frac{1}{z} - \alpha - \alpha\frac{k^{1-\alpha}(1)}{2}z - \alpha\frac{k^{1-\alpha}(2)}{3} - \dots = \frac{1}{z}\left(1 - \alpha\sum_{j=0}^{\infty}\frac{k^{1-\alpha}(j)}{j+1}z^{j+1}\right) \\
&= \frac{1}{z}\left(1 - \alpha\int\sum_{j=0}^{\infty}k^{1-\alpha}(j)z^j\,dz\right) \underset{(*)}{=} \frac{1}{z}\left(1 - \alpha\int\frac{1}{(1-z)^{1-\alpha}}\,dz\right) \\
&= \frac{1}{z}\left(1 + (1-z)^\alpha + C\right),
\end{aligned}$$

where we have used the generation formula (see [9]) in $(*)$ and C denotes an integration constant. Finally, equalling that expression to the symbol of Δ^α as a Toeplitz operator, we have

$$\Phi(z) = \frac{1}{z}\left(1 + (1-z)^\alpha + C\right) = \frac{(1-z)^\alpha}{z},$$

which implies $C = -1$.

■

The next theorem [10] shows that Δ^α is a well-defined Toeplitz operator on $\ell^2(\mathbb{N}_0)$ and exhibits chaos for any $0 < \alpha < 1$.

Theorem 21. *For any $0 < \alpha < 1$, the operator Δ^α defines a Devaney chaotic Toeplitz operator on $\ell^2(\mathbb{N}_0)$ with symbol $\Phi(z) = \frac{(1-z)^\alpha}{z}$.*

Proof. We have already seen that Δ^α is a Toeplitz operator with symbol $\dfrac{(1-z)^\alpha}{z}$ in Lemma 20. Therefore we have to prove that the operator is well-defined on $\ell^2(\mathbb{N}_0)$ and study its chaotic behavior.

1. For $0 < \alpha < 1$ and $u \in \ell^2(\mathbb{N}_0)$:
$$\Delta^\alpha u = \Delta(k^{1-\alpha} * u) = \Delta k^{1-\alpha} * u + \tau_1 u,$$
where τ_1 denotes the translation operator. Using Young's convolution inequality (Theorem 5):
$$\|\Delta^\alpha u\|_2 \leq \|\Delta k^{1-\alpha} * u\|_2 + \|u\|_2 \leq \|\Delta k^{1-\alpha}\|_1 \|u\|_2 + \|u\|_2,$$
where $\Delta k^{1-\alpha}(n) \sim \frac{C}{n^{\alpha+1}}$ by [6].
2. Using the criterion by Ganigi and Uralegaddi (Theorem 4) for univalence when $n = 0$, given $z = a+ib$ with $z \in \mathbb{D}$, we have:
$$\begin{aligned} \Re\left(\frac{D^1\Phi(z)}{\Phi(z)} - 2\right) &= \Re\left(-1 - \frac{\alpha z}{1-z}\right) = \Re\left(-1 - \frac{\alpha(a+ib)}{1-a-ib}\right) \\ &= \Re\left(-1 - \frac{\alpha(a+ib)(1-a+ib)}{(1-a)^2+b^2}\right) \\ &= \frac{-(1-a)^2 - b^2 - \alpha(1-a)a - b^2}{(1-a)^2+b^2} \\ &= \frac{(1-a)(-1+a(1-\alpha)) - 2b^2}{(1-a)^2+b^2} < 0. \end{aligned} \tag{7}$$
We have to see that $-1 + a(1-\alpha) < 0$, but $a < 1$ and $1 - \alpha < 1$, then, (7) is < 0.
3. By Baranov and Lishanskii (Theorem 3), we will show that $[-2^\alpha, 0] \subset \mathbb{C}\backslash\Phi(\mathbb{D})$. Taking the parametrization $z = e^{it}$, we obtain:
$$\Phi(e^{it}) = \frac{(1-e^{it})^\alpha}{e^{it}} = 2^\alpha \sin(t/2)^\alpha e^{i(-\alpha\pi/2+(\alpha/2-1)t)}.$$
In Figure 1 we can see that $\Phi(\mathbb{D})$ is the orange region because $\Phi(0) = \infty$. On the other hand, taking $t = 0$ and $t = \pi$ in the parametrization, we have: $\Phi(e^0) = 0$ and $\Phi(e^{i\pi}) = -2^\alpha$; therefore, the points -2^α and 0 are on the boundary of $\Phi(\mathbb{D})$, i.e., the boundary of the cardioid. So, $\{-2^\alpha, 0\} \in \mathbb{C}\backslash\Phi(\mathbb{D})$ which implies $[-2^\alpha, 0] \subset \mathbb{C}\backslash\Phi(\mathbb{D})$.

■

A very interesting consequence of this theorem when combined with the transference principle (Theorem 19) is the following result [10].

Corollary 22. *For any $0 < \alpha \leq 1$ and $a > 0$, the nabla fractional difference operator ∇_a^α is chaotic in $\ell^2(\mathbb{N}_a)$.*

Proof. We will deal with the case $\alpha = 1$ later (see Corollary 25). Using the Transference principle (Theorem 19) it follows that:
$$\tau_a \circ \nabla_a^\alpha = \Delta^\alpha \circ \tau_a.$$
We also know that Devaney chaos is preserved under quasi-conjugacy and we have just proved that the fractional difference operator is Devaney chaotic in $\ell^2(\mathbb{N}_a)$ (Theorem 21), so does ∇_a^α.

■

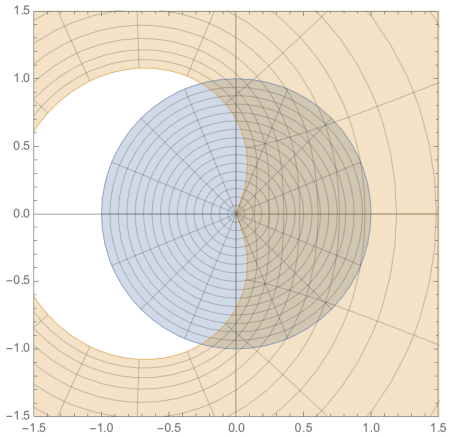

Figure 1: In orange $\Phi(\mathbb{D})$, $\alpha = \frac{3}{4}$.

4. Chaos of operators associated to Numerical Schemes

In this section we will study the chaos for relevant nonlocal difference operators that arise after analysing time-stepping numerical schemes for fractional operators. First, we consider the fractional evolution equation for $0 < \alpha < 1$

$$\partial_t^\alpha u(t) = Au(t) + f(t), \quad t > 0,$$

with initial conditions $u(0) = 0$ and ∂_t^α denoting the Riemann-Liouville fractional derivative:

$$\partial_t^\alpha u(t) = \frac{1}{\Gamma(n-\alpha)} \frac{\partial^n}{\partial t^n} \int_0^t (t-s)^{n-\alpha-1} u(s) ds, \quad n-1 < \alpha < n, n \in \mathbb{N}.$$

Time-stepping schemes for fractional operators are defined by a convolution operator $\partial_b^\alpha : \ell^2(\mathbb{N}_0) \to \ell^2(\mathbb{N}_0)$:

$$\partial_b^\alpha u(n) := (b * u)(n) = \sum_{j=0}^{n} b(n-j)u(j), \quad n \in \mathbb{N}_0, b \in \ell^1(\mathbb{N}_0).$$

Each scheme is uniquely determined by the generating series, that is the Gelfand transform of b,

$$\delta(\xi) := \sum_{n=0}^{\infty} b(n)\xi^n, \quad \xi \in \mathbb{T},$$

where $\delta(\xi)$ is called the symbol of the scheme.

The operators which we are going to study the chaos for are the adjoint operators of ∂_b^α in $\ell^2(\mathbb{N}_0)$. We remind that the adjoint operator, $(\partial_b^\alpha)^* u$, is defined as the only one which satisfies $\left\langle (\partial_b^\alpha)^* u, v \right\rangle = \langle u, \partial_b^\alpha v \rangle$. The following lemma which gives an expression for the adjoint operator of δ_b^α will be very useful.

Lemma 23. *Let $\partial_b^\alpha : \ell^2(\mathbb{N}_0) \to \ell^2(\mathbb{N}_0)$ defined, as before, by a convolution operator:*

$$\partial_b^\alpha u(n) := \sum_{j=0}^{n} b(n-j)u(j), \quad n \in \mathbb{N}_0, b \in \ell^1(\mathbb{N}_0).$$

In this case the adjoint operator, $(\partial_b^\alpha)^* u$, takes the form

$$(\partial_b^\alpha)^* u(n) = \sum_{j=0}^{\infty} b(j) B^j u(n), \quad n \in \mathbb{N}_0,$$

where B denotes the backward operator (1).

Proof.

$$\begin{aligned}\langle u, \partial_b^\alpha v\rangle &= \sum_{n=0}^{\infty} u(n)\partial^\alpha v(n) = \sum_{n=0}^{\infty}\sum_{i=0}^{n} u(n)b(n-i)v(i) = \sum_{i=0}^{\infty}\sum_{n=i}^{\infty} u(n)b(n-i)v(i)\\ &= \sum_{i=0}^{\infty} v(i)\sum_{m=0}^{\infty} u(m+i)b(m) = \sum_{i=0}^{\infty} v(i)\sum_{m=0}^{\infty}(B^m u)(i)b(m)\\ &= \langle \sum_{m=0}^{\infty} b(m)(B^m u), v\rangle = \left\langle (\partial_b^\alpha)^* u, v\right\rangle.\end{aligned}$$

And using the uniqueness of the adjoint operator we can deduce the result.

■

In order to study the conditions under which these operators are chaotic, we will use the following result from [10], that provides us with an useful criterion.

Theorem 24 ([10], Theorem V.4). *Let $b \in \ell^1(\mathbb{N}_0)$ and $F_b : \ell^2(\mathbb{N}_0) \to \ell^2(\mathbb{N}_0)$ given by*

$$F_b u(n) = \sum_{j=0}^{\infty} b(j)B^j u(n), \quad n \in \mathbb{N}_0,$$

where B denotes the backward shift operator. Then F_b defines a bounded operator on $\ell^2(\mathbb{N}_0)$ and the following assertions are equivalent

(i) F_b is chaotic;

(ii) $\delta(\mathbb{D}) \cap \mathbb{T} \neq \emptyset$.

Corollary 25 ([10]). *The forward Euler operator, $\Delta u(n) = u(n+1) - u(n)$, is chaotic.*

Proof. Firstly, we can rewrite the forward Euler operator as $\Delta = B - I$. In this way, we can apply Theorem 24, with coefficients $b(0) = -1, b(1) = 1$ and the rest of them equal to 0.

We can calculate the symbol as the Gelfand transform: $\delta(z) = z - 1$.

Now we can easily use Theorem 24 as $\delta(0) = -1 \in \mathbb{T}$ and, consequently, $\delta(\mathbb{D}) \cap \mathbb{T} \neq \emptyset$. Thus, the forward Euler operator is chaotic.

■

The symbol for the fractional backward Euler scheme is given by

$$\delta(\xi) = \tau^{-\alpha}(1-\xi)^\alpha,$$

where $\tau > 0$ denotes the step size of the scheme [8]. The corresponding kernel sequence that defines this scheme is given by $b_\tau(n) = \tau^{-\alpha}k^{-\alpha}(n)$ leading to the nonlocal operator:

$$\partial_k^\alpha u(n) = \sum_{j=0}^{n} \tau^{-\alpha}k^{-\alpha}(n-j)u(j).$$

It is remarkable that $(\partial_k^\alpha)^* = W_\tau^\alpha$ corresponds to the Weil fractional difference operator of order $\alpha > 0$. In [10] the authors prove the following interesting result that shows that chaos for W_τ^α depends on the step size τ but not on the fractional order derivative in contrast with the nonlocal operator Δ^α .

Theorem 26. *For any $\alpha > 0$, the Weil difference operator W_τ^α is chaotic on $\ell^2(\mathbb{N}_0)$ if and only if $0 < \tau < 2$.*

Proof. Note that $w \in \delta_\tau(\mathbb{D}) \cap \mathbb{T}$ if and only if

$$w = \tau^{-\alpha}(1-z)^\alpha, \qquad \text{where } |w| = 1,\ |z| < 1.$$

Then, $|1-\tau w^{1/\alpha}| = |z|$ shows that the complex number $\tau w^{1/\alpha}$ must belong to the disk of center 1 and radius 1.

Consequently, $0 < \tau < 2$ iff $\delta_\tau(\mathbb{D}) \cap \mathbb{T} \neq \emptyset$.

■

Now, we consider the fractional second order backward difference scheme given by the symbol

$$\delta(\xi) = \tau^{-\alpha}\left(\frac{3}{2} - 2\xi + \frac{1}{2}\xi^2\right)^\alpha.$$

The next result [10] shows, once again, the independence of chaos from the fractional order.

Theorem 27. *Let $0 < \alpha < 1$. The operator F_b, which is the dual of the operator that defines the fractional second order backward difference scheme with step size τ, is chaotic on $\ell^2(\mathbb{N}_0)$ if and only if $0 < \tau < 4$.*

Proof. To see an alternative proof, see [10]. As before, $w \in \delta_\tau(\mathbb{D}) \cap \mathbb{T}$ if and only if

$$w = \tau^{-\alpha}\left(\frac{3}{2} - 2z + \frac{1}{2}z^2\right)^\alpha = \tau^{-\alpha}\left(\frac{3}{2}\right)^\alpha (1-z)^\alpha \left(1 - \frac{z}{3}\right)^\alpha,$$

where $|w| = 1$, $|z| < 1$. Then

$$w^{\frac{1}{\alpha}}\tau = \frac{3}{2} - 2z + \frac{1}{2}z^2.$$

And solving this second order equation we obtain

$$z - 2 = \pm\sqrt{1 + 2w^{\frac{1}{\alpha}}\tau}.$$

Geometrically, the left side is the disk of radius 1 and center -2, and the right side is the square root of the circumference of center 1 and radius 2τ. We need to see the values of τ for which they intersect (thus, the operator will be chaotic), so we must consider the minus sign of the square root.

The right hand side is, therefore, a close curve that encircles -1 (which lays at the frontier of the right-hand-side disk). So as τ increase the curve is larger, and the upper limit is $\tau = 4$ for which that curve only intersects the frontier and not the open disk. The lower bound is $\tau = 0$, since for any higher value below 4 the curve will intersect the disk as it encircles -1.

So, the operator F_b will be chaotic if and only if $0 < \tau < 4$.

■

Finally, we consider the Crank-Nicholson stepping scheme with symbol [8]:

$$\delta(\xi) = \tau^{-\alpha}\frac{(1-\xi)^\alpha}{1 - \frac{\alpha}{2} + \frac{\alpha}{2}\xi}.$$

In the case $0 < \alpha < 1$, Lizama, Murillo and Peris arrive in [10] at the following result, which reveals a surprising connection of chaos not only with the step size but also with the fractional order α.

Theorem 28. *Let $0 < \alpha < 1$. The operator F_b, which is the dual of the operator that defines the fractional Crank-Nicholson scheme with step size τ, is chaotic on $\ell^2(\mathbb{N}_0)$ if and only if $0 < \tau < \dfrac{2}{(1-\alpha)^{1/\alpha}}$.*

Proof. Again, it suffices to prove that $\delta_\tau(\mathbb{D}) \cap \mathbb{T} \neq \emptyset$ iff $0 < \tau < \dfrac{2}{(1-\alpha)^{1/\alpha}}$. By the maximum principle, $\sup_{z\in\mathbb{D}} |\delta(z)| = \max_{z\in\mathbb{T}} |\delta(z)|$, attained at -1. Moreover, $\delta(1) = 0 \in \mathbb{D}$.
Thus, $\delta_\tau(\mathbb{D}) \cap \mathbb{T} \neq \emptyset$ iff $\frac{2^\alpha}{1-\alpha}\tau^{-\alpha} > 1$ iff both $\tau > 0$ and $\frac{2}{(1-\alpha)^{1/\alpha}}\tau^{-\alpha}$ are satisfied. ■

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